



Tomato Septoria Leaf Spot

Tomato septoria leaf spot is a common fungal disease of solanaceous crops caused by *Septoria lycopersici*. Symptoms appear on the leaves as circular, tan-to-gray spots with darker brown margins and dotted with dark, raised pycnidia inside the lesion. These lesions are often surrounded by a yellow halo. The disease typically develops on lower leaves first, moving up to higher leaves as infection progresses. Lesions can converge and lead to defoliation of lower leaves, and in severe cases the death of an entire plant. Stem lesions appear similar to leaf lesions, but are often darker. Fruit lesions are uncommon, but appear similar to leaf lesions under very high disease pressure.

Primary source: Primarily diseased solanaceous crop or weed debris in soil, also infected seeds and equipment

Spread: Rainfall, irrigation, workers, equipment, and several insects

Favorable conditions: High humidity, moderate temperatures (68-77°F), high dew point/wet conditions, poor airflow

Infection & Disease Cycle

The main source of inoculum for primary infections is *S. lycopersici* spores that overwinter on diseased solanaceous crop or weed debris in the soil. The fungus can also survive on equipment, as well as infected seed, which will produce diseased seedlings. Spores (conidia) are produced and spread during wet and warm periods, especially when airflow is poor due to canopy closure and densely-spaced plants. These spores are spread from primarily debris to leaves via rainfall, irrigation, workers, equipment, and several insects, often reaching and infecting the lower and older leaves

first. These spores penetrate the leaf tissue via the stomata, leading to lesion development within ~5 days. Pycnidia (asexual fruiting bodies) will develop ~14 days after inoculation, releasing more spores that will be spread and create new, secondary infections of healthy plant tissue including leaves, stems, and fruit.

Cultural Control

Scouting regularly allows early identification of disease before significant spread and damage. Disease spread can be limited by proper mulching, which can reduce plant-soil contact, as well as disinfecting tools and equipment like stakes and cages. The following practices can also help prevent disease development:

- Plant resistant varieties
- Rotate away from susceptible solanaceous crops (1-2 years)
- Stake or trellis plants to improve airflow
- Remove 'suckers' or lowest lateral tomato plant growth (<https://hort.extension.wisc.edu/articles/tomato-pruning/>)
- Maintain proper spacing between plants
- Control host weeds
- Destroy infested plant debris
- Avoid over-irrigating (reduce leaf wetness)

Chemical Control:

Preventative applications of fungicides containing copper or chlorothalonil can be useful in areas with chronic *Septoria lycopersici* infections. For Wisconsin-specific fungicide information, refer to the annually updated [Commercial Vegetable Production in Wisconsin \(A3422\)](#). Or, for home garden fungicide recommendations, see [Home Vegetable Garden Fungicides \(D0062\)](#), a fact sheet available through the UW Plant Disease Diagnostic Clinic website. Always follow label directions carefully.

Resources

- [Commercial Vegetable Production in Wisconsin \(A3422\)](#). This guide offers the latest recommendations for disease, insect, and weed management in Wisconsin's most common commercial vegetable crops. Also included are lime and fertilizer recommendations as well as insect identification information and keys.

- [UW Plant Disease Diagnostics Clinic](#). The University of Wisconsin–Madison/Extension Plant Disease Diagnostics Clinic (PDDC) provides assistance in identifying plant diseases and provides educational information on plant diseases and their control.

References

- Douglas, Sharon M. "Septoria Leaf Spot of Tomato." The Connecticut Agricultural Experiment Station.
- Hudelson, Brian. "Septoria Leaf Spot." UW Plant Disease Diagnostics Clinic, May 22, 2021. https://pddc.wisc.edu/wp-content/blogs.dir/39/files/Fact_Sheets/FC_PDF/Septoria_Leaf_Spot.pdf.



Tomato leaf showing symptoms of Septoria leaf spot. Photo credit: Bruce Watt, University of Maine, via Bugwood.org



Septoria leaf spot symptoms on tomato leaves and stem. Photo credit: Paul Bachi, University of Kentucky Research and Education Center, Bugwood.org

For diagnostic and management support in Wisconsin, please consider contacting Dr. Amanda Gevens (gevens@wisc.edu) or our [Plant Disease Diagnostic Clinic](#) (pddc@wisc.edu).

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